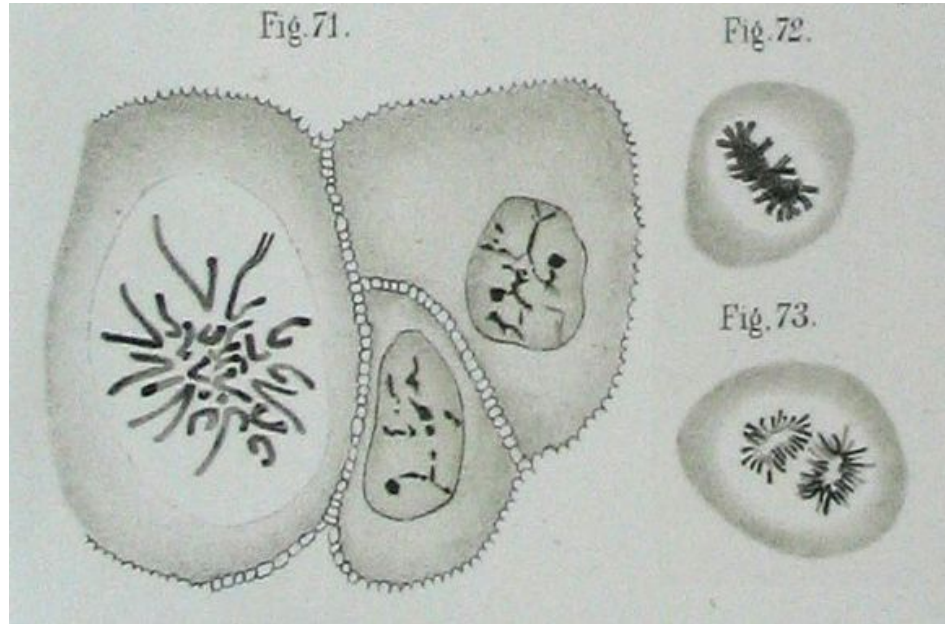


Introduction to Biology

Methods in Bioinformatics (TI) - 2026/27

Principal players: ...?



Walther Flemming, 1881

Principal players: molecules

→ Deoxyribonucleic acid (DNA)

→ Ribonucleic acid (RNA)

→ Proteins

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- ◆ Contains the genetic information passed from one generation to the next one
- ◆ Digital form: ATGCGTATAGAGTCT... (4)

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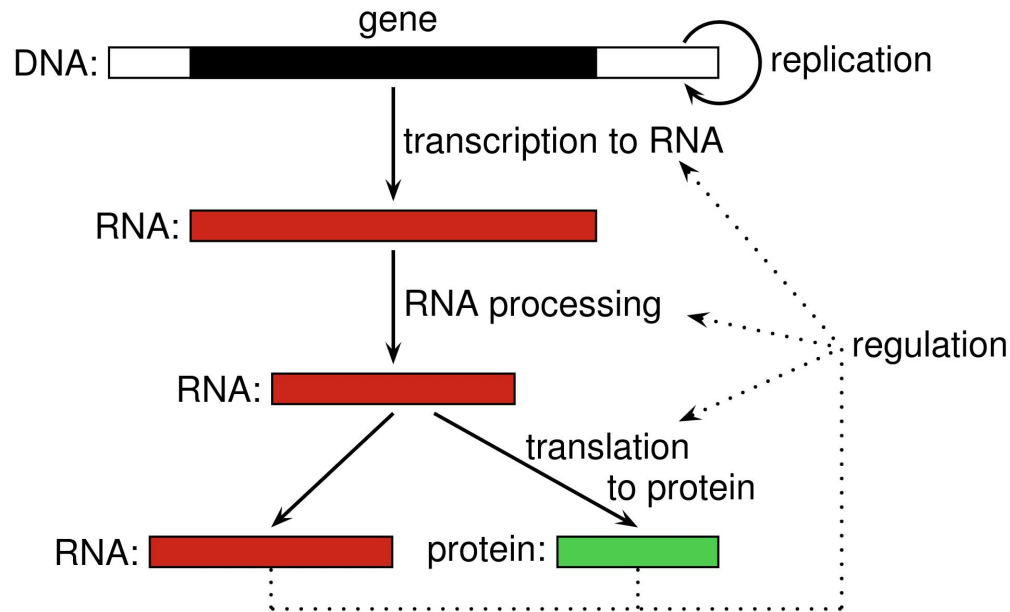
- ◆ Similar to DNA, but Thymine is replaced by Uracile

→ Proteins

- ◆ Molecules built from genetic information contained in the DNA
 - *Catalyze biochemical reactions in the cell*
 - *Carry signals within/between cells,*
 - *Important for cell structure and movement*
 - ...
- ◆ Digital form: MATHALEIAGLF... (20)

What information is stored in DNA?

DNA contains the recipes (**genes**) for the synthesis of proteins and functional RNA



! Simplification: 1 gene → 1 protein

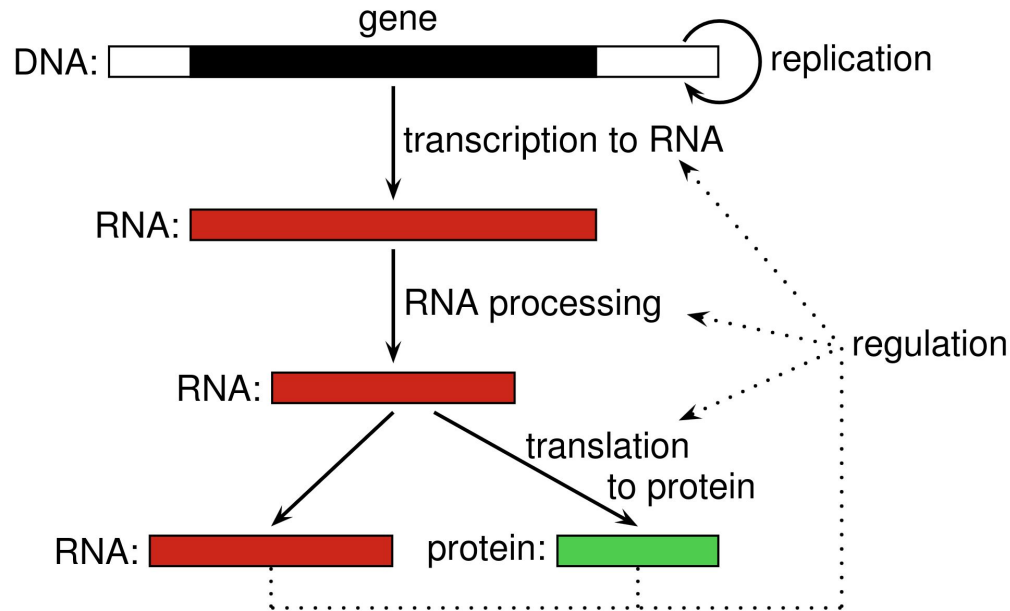
Central dogma (Francis Crick 1958,1970)

“Biological information generally flows: DNA → RNA → protein”

- Information flows from nucleic acids to proteins
- Information cannot flow from proteins back to nucleic acids
- Information cannot flow from proteins to other proteins

Regulation of gene expression (gene → protein)

When and how many molecules to synthesize (*e.g., condition, tissue, protein regulation...*)



DNA

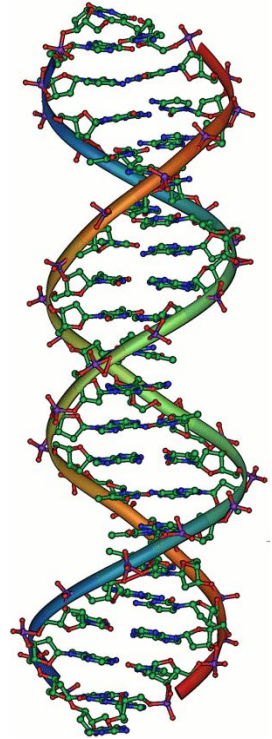
→ Two complementary strands (A/T, C/G) in opposite orientation (5'/3')

ACCATG → * CATGGT

→ Double stranded structure:

- ◆ Redundancy (*if one strand is damaged, it can be fixed*)
- ◆ Easy replication during cell division (*DNA unwinds and two copies are created*)

→ The **genome** is all DNA contained in a cell, usually split in **chromosomes**



Human genome has 22 pairs of chromosomes plus two sex chromosomes, together 3.1GB (billion base pairs)

**Reverse-and-complement (source of many bugs)*

DNA sequencing:

- Technology for determining sequence of nucleotides in chromosomes
- Complex process that requires multiple laboratory steps.
 1. Chromosome are cut in short pieces
 2. Each piece is duplicated many times
 3. Each piece is sequenced separately, e.g., by Sanger sequencing (*oldest, uses natural enzymes*)

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DNA polymerase:

- builds a new DNA strand from an existing DNA strand
- extend a shorter strand from 5' to 3' using the other as template

Sanger Sequencing

Determines the order of DNA bases by creating DNA fragments stopping at specific bases

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Determines the order of DNA bases by creating DNA fragments stopping at specific bases

We need in a test tube:

- ✓ Many copies of the DNA sample (*single-stranded*)
- ✓ **Primers** (*to start polymerase*)
- ✓ Two types of nucleotides:
 - i. Normal
 - ii. Color-coded and modified to block polymerase



Sanger Sequencing

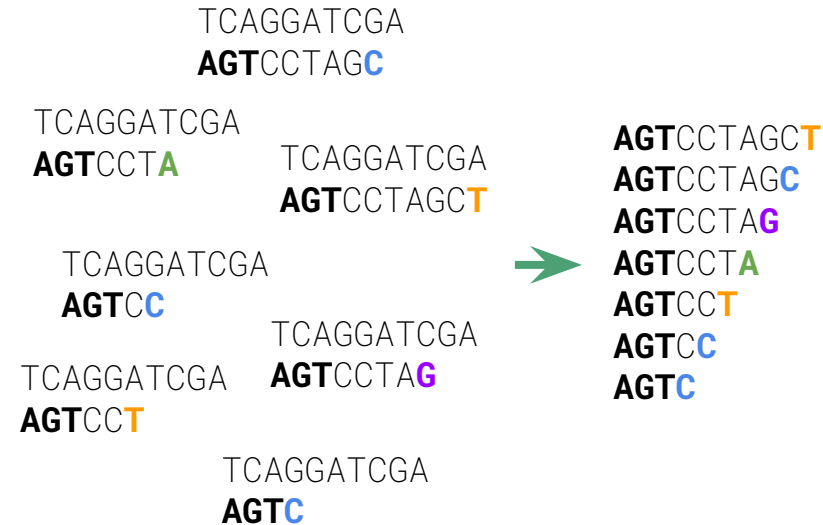
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We need in a test tube:

- ✓ Many copies of the DNA sample (*single-stranded*)
- ✓ **Primers** (*to start polymerase*)
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 - i. Normal
 - ii. Color-coded and modified to block polymerase

Steps:

1. Binding via hybridization
2. DNA polymerase
3. Strand separation and size sorting
4. DNA reconstruction* using colors



**we have so many that we are "sure" to cover each base*

DNA sequencing issues

- ✗ DNA cannot be read as a single long fragment, but in many short* pieces (***reads***)
- ✗ DNA cannot be always read correctly (error rate*)

**vary by technology*

DNA sequencing issues

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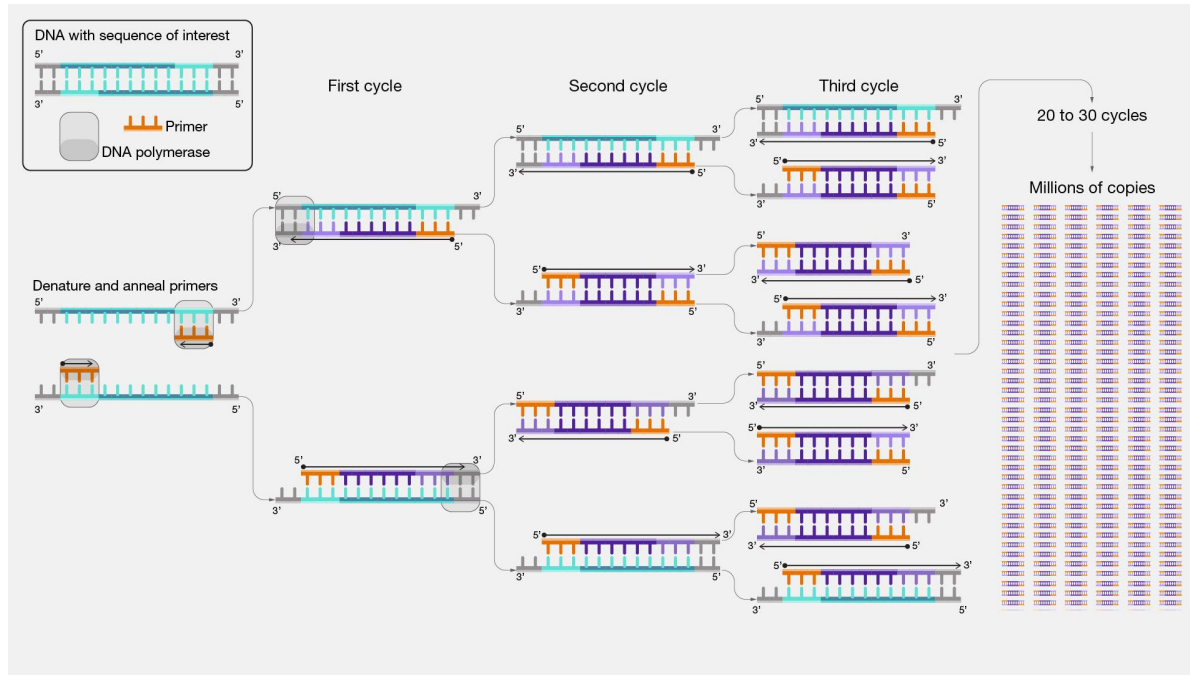
Computational problems:

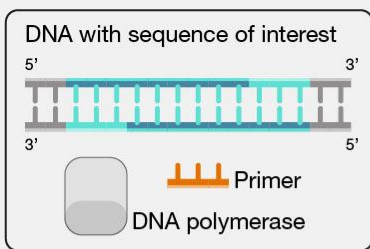
- Genome assembly
 - ↳ Annotate genes and other functional regions
 - ↳ Seek similarities and differences between species and organisms
 - ↳ Read alignment

**vary by technology*

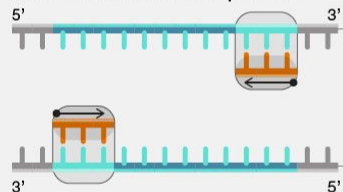
PCR (Polymerase Chain Reaction)

- Mechanisms to duplicate DNA
- Also useful for testing (e.g., SARS-CoV-2)

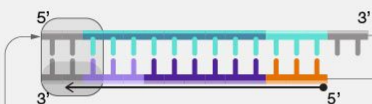




Denature and anneal primers



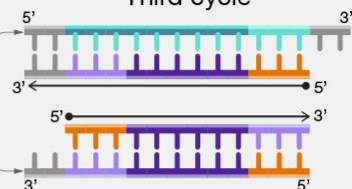
First cycle



Second cycle

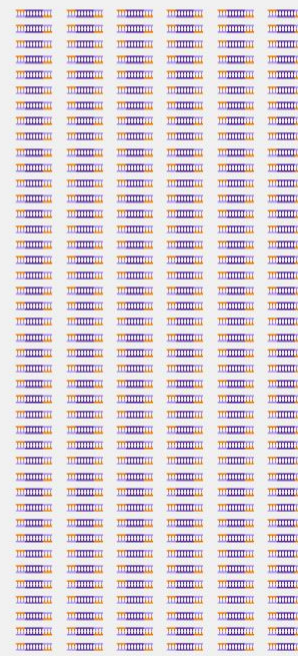


Third cycle



20 to 30 cycles

Millions of copies

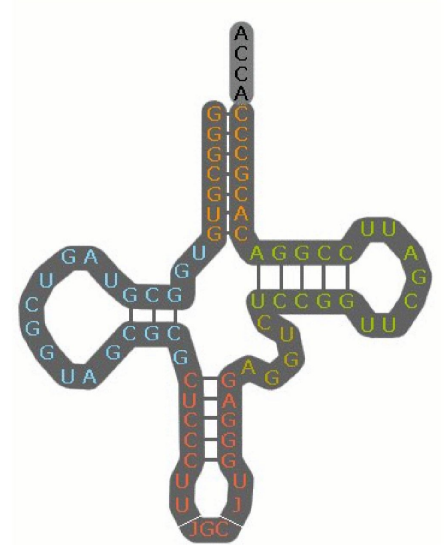


1. Heat to separate
2. Cool to bind (👉)
3. DNA polymerase
4. Repeat (exponential process)

RNA - How proteins are made

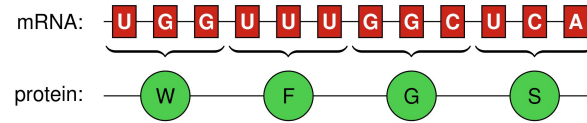
Differently from DNA, RNA:

- Single-stranded sequences, and usually shorter
- Ribose instead of deoxyribose
- Uracil instead of thymine (bases A,C,G,U)
- Complex secondary structure (*how strand fold due to paired complementary regions*)



Translation: from RNA to protein

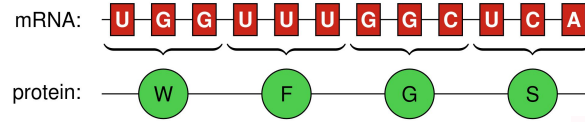
From triplets of {A,C,G,U} to 20 single letter amino acids



→ How many codons/triplets?

Translation: from RNA to protein

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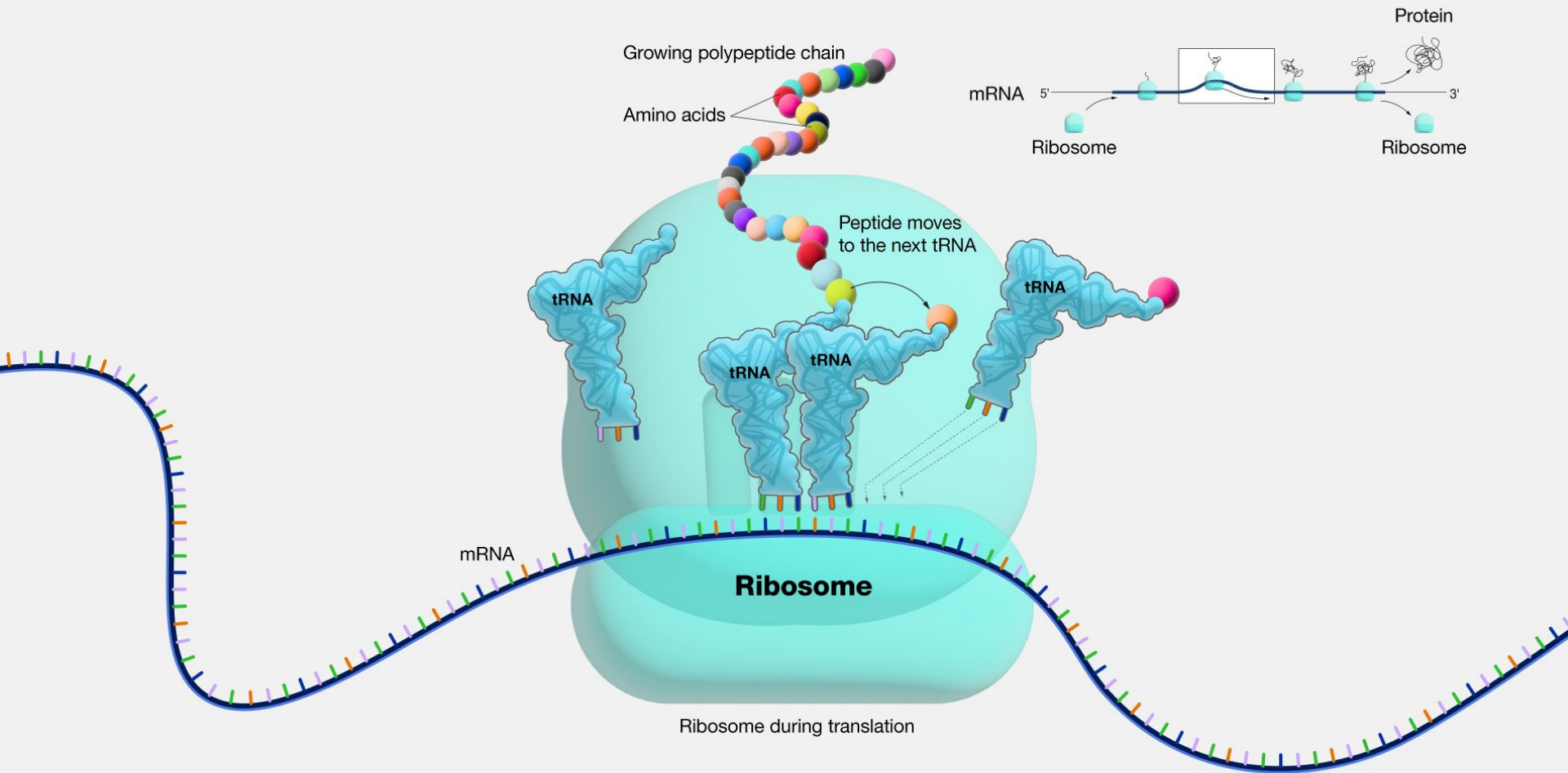


→ $4^3 = 64 \gg 20$ (not a 1-to-1 correspondence)

Codon Chart

		Second base in codon				
		U	C	A	G	
First base in codon	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } STOP UAG }	UGU } Cys UGC } UGA } STOP UGG } Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } Arg CGC } CGA } CGG }	U C A G
	A	AUU } Ile AUA } AUG } Met (start)	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } Val GUC } GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } Gly GGC } GGA } GGG }	U C A G

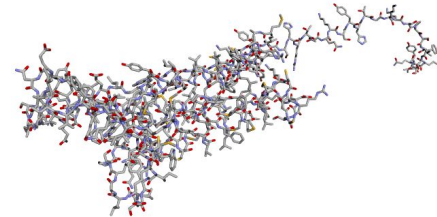
Image from <https://microbenotes.com/codon-chart-table-amino-acids/>



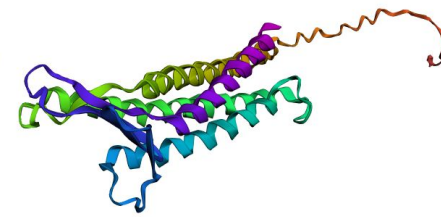
Protein structure

Proteins are sequences of amino acids but are organized in very precise shapes

- Proteins occur **folded** in a stable 3D structures, that mainly depends on amino acids
 - ◆ Each amino acid has a specific structure and specific chemical properties
 - *hydrophobic/hydrophilic*
 - *positive/negative charge*
- Structure of a protein determines its function
- Proteins can switch between conformations

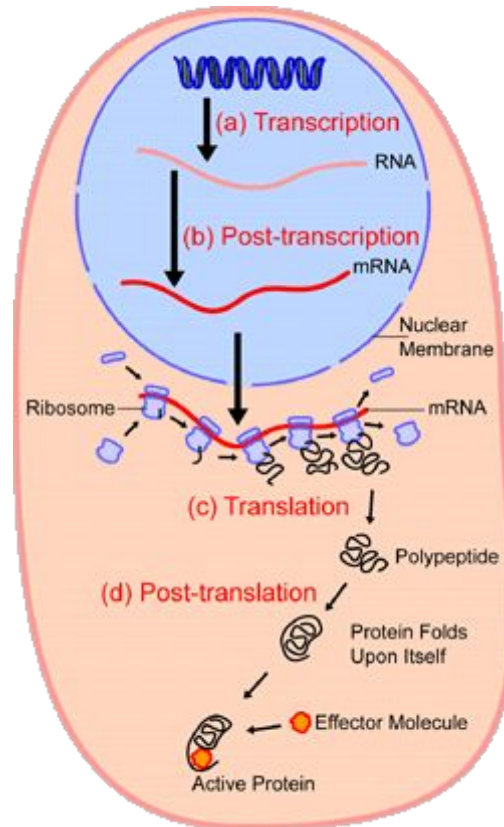


Atomic structure



Tertiary structure

Overview

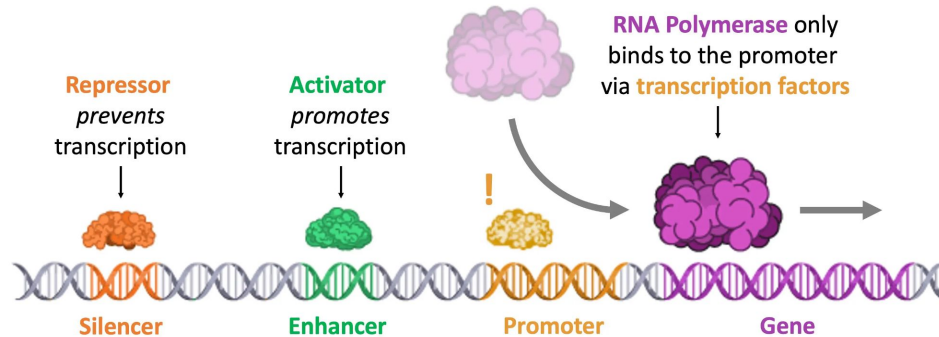


Regulation of gene expression

- Cells share the same genome, yet look and function differently
 - ◆ Some proteins are produced only under special circumstances or in variable amounts

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Computational problems:

- How are different proteins regulated in a cell?
- Which factors influence a given gene?
- Where they bind?

How to find out which genes are active

→ **Microarray:** technology to measure expression levels of genes under 1/+ conditions

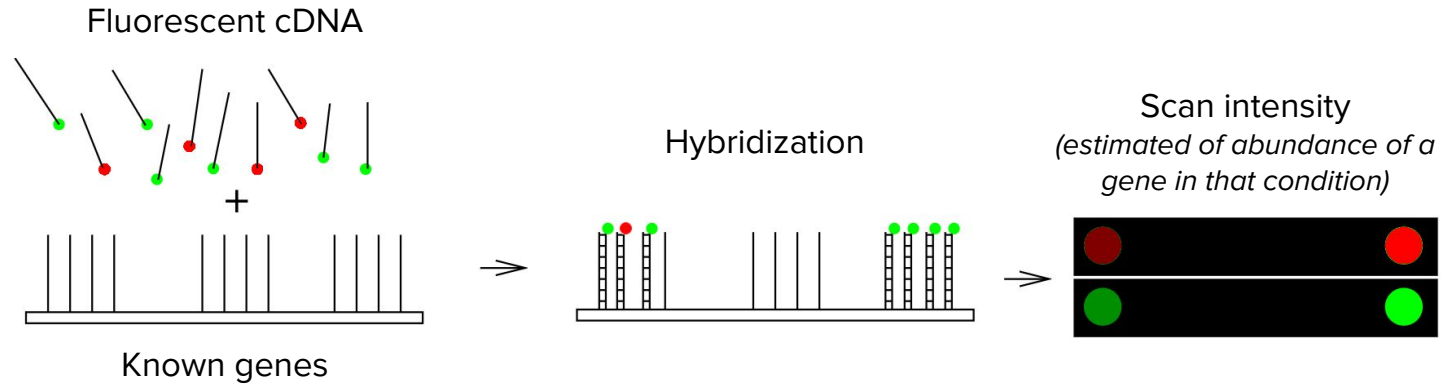
- ◆ Typically used to compare conditions:
 - Healthy versus diseased tissue
 - Treated versus untreated cells
 - Different developmental stages



Does not measure protein production

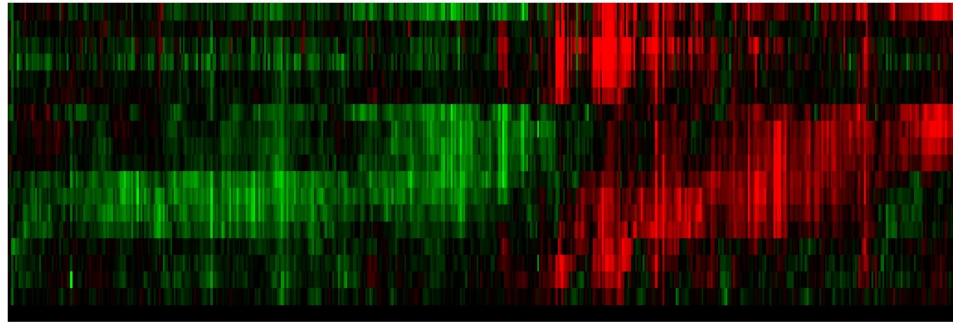
- ◆ if a gene is expressed, it doesn't mean the corresponding protein is being produced

Microarray



Example of microarray data

Two conditions (red/green), 517 genes, 19 time points



19 experiments

517 genes



Computational problem:

→ determine which genes correlates together?

Mutations in DNA

- Occasionally, DNA mutates (*e.g., environmental factors, damage, replication errors...*)
- Three main mutation types:
 - ◆ Single-Nucleotide Variants (SNVs)
 - ◆ Short indels
 - ◆ Structural Variants (SVs)

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Computational problems:

- *Variant calling*
- *Homology search (e.g., highly conserved mutations)*
- *Sequence alignment*

Populations genetics

“Study of mutations in populations over generations”

→ Mutations occur in a particular cell at a particular time...

→ ...and are then propagated from parents to offspring



Dangerous mutations (*quickly eliminated*)



Advantageous mutations (*more likely to spread in the population, natural selection*)

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Computational problem:

- determine polymorphisms linked to a certain phenotype

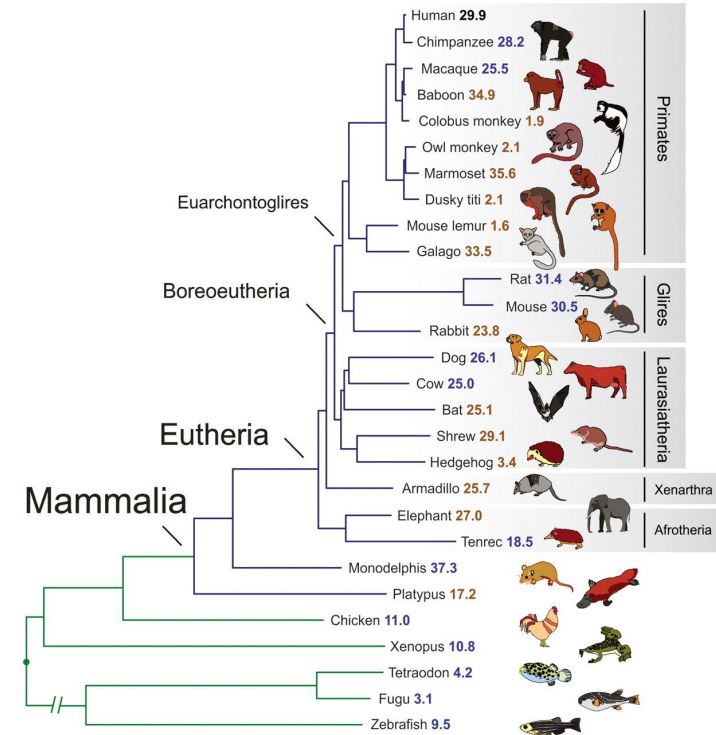
Evolution

Study of DNA modifications over loooooong periods of time
(*speciation*: “splitting of species”)



Computational problem:

→ given DNA of current species, reconstruct their evolutionary history (**phylogenetic tree**)



Type of organisms

→ **Prokaryotes** (*bacteria and archaea*)

- ◆ Simple unicellular organisms
- ◆ DNA directly in the cytoplasm
- ◆ Genome in one circular chromosome, plus shorter plasmids
- ◆ Simple gene structure

→ **Eukaryotes** (*animals, fungi, plants, some unicellular organisms*)

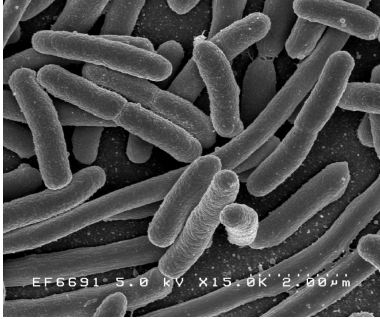
- ◆ DNA in a nucleus, cell contains also other organelles
- ◆ Longer genome in several linear chromosomes
- ◆ Mitochondria/chloroplasts are “engulfed prokaryotes”

Model organisms

Species important for biology research that are studied more than others

- Ideal to work with (simple to grow and/or reproduce)
- Easy to experiment with
- Discoveries can be transferred to other species

Model organisms (II)



Escherichia coli:

- Bacterium living in digestive tract
- Simple to manipulate: cell division every 20 min
- Study of basic biological processes: DNA replication, gene expression...
- Genome 4.6MB, 4000 genes.



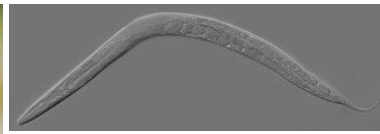
Saccharomyces cerevisiae:

- Used to make bread
- Simple eukaryotic organism (genome with 6000 genes, 13MB)
- Cell division every 2 hours
- Study of processes specific for eukaryotes

Model organisms (III)

Arabidopsis thaliana:

- Small flowering plant
- 6-week reproduction cycle
- Model organism for plant research



Invertebrates:

- Caenorhabditis elegans (worm) - *study of development, cell differentiation*
- Drosophila melanogaster (fruit fly) - *study of genetics, development genes*

Vertebrates:

- Xenopus laevis (frog) - *large eggs, easy to manipulate*
- Danio rerio (zebrafish) - *translucent embryos*
- Mus musculus (mouse) - *many laboratory breeds with different properties*



Available data

- Most of the data is freely available
 - ◆ DNA sequences
 - ◆ Read samples
 - ◆ Genome annotation (location of genes and other functional elements)
 - ◆ RNA sequences and structures
 - ◆ Protein sequences, their function and structure
 - ◆ Measurements of cell state (amount of RNA, protein...)
- Data are obtained via experiments and/or by computational methods
 - ! Often unreliable, noisy (**in both cases**)

More information

- ✓ Zvelebil, Baum: Understanding Bioinformatics, chapter 1
- ✓ University textbooks of molecular biology
- ✓ English wikipedia
- ✓ Tutorials linked on the course website